



**Generation Interconnection
System Impact Study Report
for**

Queue Project AF2-082

DAUPHIN PG TIEDAUPHIN JUNIATA TIE 69 K

33 MW Capacity / 55 MW Energy

February 2021

Table of Contents

1	Introduction.....	3
2	Preface.....	3
3	General	4
4	Point of Interconnection.....	4
5	Cost Summary	4
6	Transmission Owner Scope of Work	5
6.1	Attachment Facilities Cost Estimate	5
6.2	Direct Connection Cost Estimate.....	6
6.3	Non-Direct Connection Cost Estimate.....	6
7	Schedule.....	7
8	Interconnection Customer Requirements.....	8
9	Revenue Metering and SCADA Requirements	8
9.1	PJM Requirements	8
9.2	Meteorological Data Reporting Requirements	8
9.3	Interconnected Transmission Owner Requirements.....	8
10	Summer Peak Analysis	8
10.1	Generation Deliverability	8
10.2	Multiple Facility Contingency	9
10.3	Contribution to Previously Identified Overloads.....	9
10.4	Steady-State Voltage Requirements.....	9
10.5	Potential Congestion due to Local Energy Deliverability.....	9
10.6	System Reinforcements.....	9
11	Light Load Analysis.....	9
12	Short Circuit.....	9
12.1	System Reinforcements - Short Circuit.....	9
13	Stability and Reactive Power	10
14	Affected Systems	10
14.1	NYISO	10
15	Attachment 1: One Line Diagram	11

1 Introduction

This System Impact Study has been prepared in accordance with the PJM Open Access Transmission Tariff, 205, as well as the System Impact Study Agreement between the Interconnection Customer (IC), and PJM Interconnection, LLC (PJM), Transmission Provider (TP). The Interconnected Transmission Owner (ITO) is PPL.

2 Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

The Interconnection Customer seeking to interconnect a wind or solar generation facility shall maintain meteorological data facilities as well as provide that meteorological data which is required per Schedule H to the Interconnection Service Agreement and Section 8 of Manual 14D.

3 General

The Interconnection Customer (IC), Dauphin Solar I, LLC, has proposed a Solar generating facility located in Dauphin County, Pennsylvania. The installed facilities will have a total capability of 55 MW with 33 MW of this output being recognized by PJM as Capacity. The proposed in-service date for this project is December 01, 2022. This study does not imply a TO commitment to this in-service date.

Queue Number	AF2-082
Project Name	DAUPHIN PG TIEDAUPHIN JUNIATA TIE 69 kV
State	Pennsylvania
County	Dauphin
Transmission Owner	PPL
MFO	55
MWE	55
MWC	33
Fuel	Solar
Basecase Study Year	2023

Any new service customers who can feasibly be commercially operable prior to June 1st of the basecase study year are required to request interim deliverability analysis.

4 Point of Interconnection

AF2-082 will interconnect with the PPL on transmission system tapping between the Dauphin PG Tie 69 to Dauphin Juniata Tie 69 kV buses on the Sunbury to Dauphin 69 kV line. The Point of Interconnection (POI) will be where the PPL EU tap line terminates (with insulators) at the first dead-end structure inside the IC substation.

5 Cost Summary

The AF2-082 project will be responsible for the following costs:

Description	Total Cost
Total Physical Interconnection Costs	\$ 1,555,400
Allocation towards System Network Upgrade Costs*	\$ 0
Total Costs	\$ 1,555,400

*As your project progresses through the study process and other projects modify their request or withdraw, then your cost allocation could change.

This cost excludes a Federal Income Tax Gross Up charges. This tax may or may not be charged based on whether this project meets the eligibility requirements of IRS Notice 88-129. If at a future date it is determined that the Federal Income Tax Gross charge is required, the Transmission Owner shall be reimbursed by the Interconnection Customer for such taxes.

Note 1: PJM Open Access Transmission Tariff (OATT) section 217.3A outline cost allocation rules. The rules are further clarified in PJM Manual 14A Attachment B. The allocation of costs for a network upgrade will start with the first Queue project to cause the need for the upgrade. Later queue projects will receive cost allocation contingent on their contribution to the violation and are allocated to the queues that have not closed less than 5 years following the execution of the first Interconnection Service Agreement which identifies the need for this upgrade.

6 Transmission Owner Scope of Work

The total physical interconnection costs are given in the table below:

Description	Total Cost
Attachment Facilities	\$ 893,800
Direct Connection Network Upgrade	\$ 0
Non-Direct Connection Network Upgrades	\$ 661,600
Total Physical Interconnection Costs	\$ 1,555,400

PPL EU can accommodate this interconnection by constructing a new transmission line tap off the Sunbury – Dauphin 69 kV line and completing associated remote end relay work.

Risks and Assumptions

- No major environmental, real estate, or permitting issues.
- IC is responsible for acquisition of easements, permits, and right of way for any Direct Connection Network Upgrades and Attachment Facilities per PPL EU standards and requirements.
- PPL EU will perform all grading, site preparation, and establish access roads for the PPL EU owned Attachment Facilities per PPL EU standards and requirements.

6.1 Attachment Facilities Cost Estimate

69 kV Transmission Line Tap

PPL EU will tap the Sunbury – Dauphin 69 kV line at or near GPS Coordinates: 40.5820510, -76.9076980. PPL EU will extend the tap southwest towards the IC site. PPL EU will install a motor operated switch and POI termination structure. The IC must build the remainder of the Attachment Facilities from the POI

termination structure to the IC substation. The IC is responsible for procuring 100 ft. ROW for these facilities. For the purposes of this Feasibility Study Report cost estimate, PPL EU is assuming all engineering and construction responsibility for land development activities, including grading, site preparation, and new access road. During the Facilities Study phase, PPL EU and the IC will review land development activities, and the IC may choose to perform some, or all, of these activities. The cost estimate will be updated accordingly and included in the Facilities Study Report.

PPL EU work will consist of installing the following:

- Install one (1) new single circuit, tension, custom steel/foundation, tap pole structure.
- Install one (1) new single circuit, direct-embed, steel, motor operated switch structure
- Install one (1) new single circuit, direct-embed, steel, tension structure to the dead-end inside the substation.
- New circuit will consist of three (3) phase 556.6 kcmil 24/7 ACSR conductor from the tap pole to the substation dead-end and 144-ct optical ground wire (OPGW).

The total preliminary cost estimate for the Attachment work is given in the table below:

Description	Total Cost
69 kV Transmission Line Tap	\$ 869,400
IC substation facility acceptance, testing, commissioning, & telemetry coordination	\$ 24,400
Total Attachment Facility Costs	\$ 893,800

6.2 Direct Connection Cost Estimate

None.

6.3 Non-Direct Connection Cost Estimate

Remote End Relay Work – Sunbury 69 kV Substation

- Complete remote end relay work at Sunbury 69 kV substation for Direct Transfer Trip.
- Model IC in CAPE and conduct a wide area short circuit study two busses away from the IC facilities. Identify affected relays and revise settings as needed.
- Conduct a review of the IC relay settings and engineering package (submitted by IC to PPL EU).
- PPL EU currently has OPGW on the Sunbury – Dauphin 69 kV line available as the communication circuit for DTT to the Sunbury 69 kV substation.

Remote End Relay Work – Dauphin 69 kV Substation

- Complete remote end relay work at the Dauphin 69 kV substation for Direct Transfer Trip.
- Model IC in CAPE and conduct a wide area short circuit study two busses away from the IC facilities. Identify affected relays and revise settings as needed.
- Conduct a review of the IC relay settings and engineering package (submitted by IC to PPL EU).

Optional: The Dauphin 69 kV substation is an alternate transmission source for the IC and may be utilized to radially source this project's POI during outages.

Sunbury – Dauphin 69 kV Modifications to tie in the AF2-082 Attachment Facilities

- Tie the new AF2-082 Attachment Facilities into the Sunbury - Dauphin 69 kV line.
- Remove (1) structure on the 746 – Sunbury – Dauphin 69 kV line at 23379S45582.
- Break one (1) fiber and complete a section throw onto the new substation by installing two (2) new fiber splice boxes on structures 23343S45538 and the new tap pole.
- Install motor operated switch structure on each side of the new tap.

The total preliminary cost estimate for the Non-Direct Connection work is given in the table below.

Description	Total Cost
Sunbury – Dauphin 69 kV Modifications to tie in the AF2- 082 Attachment Facilities	\$ 96,600
Install Motor Operated Switches on Sunbury – Dauphin 69 kV line, one on each side of new tap	\$ 289,000
Remote End Relay Work – Sunbury 69 kV Substation	\$138,000
Remote End Relay Work – Dauphin 69 kV Substation	\$138,000
Total Non-Direct Connection Facility Costs	\$ 661,600

7 Schedule

The estimated time to complete the scope of work is 12-18 months after the PJM three-party Interconnection Service Agreement (ISA) and Interconnection Construction Service Agreement (ICSA) are signed and PPL EU receives Notice to Proceed from the IC.

8 Interconnection Customer Requirements

PPL EU applicable technical standards that address requirements for interconnection of generation, transmission, and end user facilities can be found at the following link:

<https://pjm.com/planning/design-engineering/to-tech-standards/private-ppl.aspx>

9 Revenue Metering and SCADA Requirements

9.1 PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Section 8 of Attachment O.

9.2 Meteorological Data Reporting Requirements

The solar generation facility shall provide the Transmission Provider with site-specific meteorological data including:

- Temperature (degrees Fahrenheit)
- Atmospheric pressure (hectopascals)
- Irradiance
- Forced outage data

9.3 Interconnected Transmission Owner Requirements

The IC will be required to comply with all Interconnected Transmission Owner's revenue metering requirements for generation interconnection customers located at the following link:

<http://www.pjm.com/planning/design-engineering/to-tech-standards/>

10 Summer Peak Analysis

The Queue Project AF2-082 was evaluated as a 55 MW (Capacity 33 MW) injection tapping the Dauphin PG Tie 69 to Dauphin Juniata Tie 69 kV line in the PPL area. Project AF2-082 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AF2-082 was studied with a commercial probability of 100.0 %. Potential network impacts were as follows:

10.1 Generation Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

10.2 Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

10.3 Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

None

10.4 Steady-State Voltage Requirements

To be determined during the Facilities Study

10.5 Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

None

10.6 System Reinforcements

None

11 Light Load Analysis

Not Required

12 Short Circuit

The following Breakers are overdutied:

None

12.1 System Reinforcements - Short Circuit

None

13 Stability and Reactive Power

(Summary of the VAR requirements based upon the results of the dynamic studies)

To be determined in the Facilities Study Phase.

14 Affected Systems

14.1 NYISO

NYISO Impacts to be determined during later study phases (as applicable).

15 Attachment 1: One Line Diagram

